

Task 1 to 5

Task 1 – Adversity Profile Analysis

The adversity profile measures the percentage of trades that were adverse to the liquidity provider (LP) at different horizons $\tau \in \{5, 10, 15, 20, 25, 30\}$ seconds. A trade is classified as adverse if the realized PnL from the LP's perspective is negative when the position is closed at the specified horizon.

The results show a clear ranking of clients by toxicity. Client A exhibits the lowest adversity profile, with adversity remaining around 40–42% across all horizons. Client B displays a similar pattern but at slightly higher levels. Clients C and D show a gradual increase in adversity as the horizon increases. Clients E and F demonstrate the strongest adverse behavior, with Client F reaching approximately 62% adversity at the 30-second horizon.

A notable observation is that adversity generally increases with the holding horizon. This suggests that the information contained in client trades is incorporated into prices progressively rather than instantaneously. Clients whose adversity rises significantly over time are likely trading on information that predicts future price movements. From the LP's perspective, such clients represent more toxic order flow because adverse price movement materializes after the trade has already been executed.

Based on the adversity profiles, the clients can be ranked from least toxic to most toxic as:

$A < B < C < D < E < F$

This ranking indicates increasing levels of informed trading behavior and adverse selection risk faced by the liquidity provider.

Task 2 – Client Profitability and Spread Recommendation

Expected PnL was computed for each client at all horizons and aggregated using the uniform weighting scheme specified in the challenge. Clients A, B, C, and D exhibit positive aggregate expected PnL from the LP's perspective and are therefore classified as profitable clients. Clients E and F exhibit negative aggregate expected PnL and are classified as costly clients.

The profitability classification aligns closely with the adversity profiles observed in Task 1. Clients with lower adversity rates generate positive expected PnL for the LP, while clients with higher adversity rates generate losses. In particular, Clients E and F exhibit the highest adversity percentages and are the only clients producing negative aggregate PnL.

The recommended minimum half-spread δ^* represents the smallest spread adjustment required to make the expected aggregate PnL non-negative. For Clients A, B, and C, no additional spread is required because their aggregate PnL is already positive. Client D requires only a small adjustment, while Clients E and F require substantially larger spread compensation.

The required half-spreads are:

Client A: 0.0000

Client B: 0.0000

Client C: 0.0000

Client D: 0.0096

Client E: 0.0190

Client F: 0.0329

These results demonstrate a strong relationship between adversity and spread compensation. More adverse clients require wider spreads to offset the losses generated by adverse selection. Therefore, the adversity profile can be viewed as a direct indicator of the spread premium that should be charged by the liquidity provider.

Task 3 – Adversity Prediction Model

The objective of Task 3 is to estimate the probability that an incoming trade will be adverse at a given horizon $\tau \in \{5, 10, 15, 20, 25, 30\}$. A separate XGBoost classifier was trained for each horizon.

The feature set consisted of:

1. Side
2. Volume
3. Trade Price
4. M0 (current mid-price)
5. Spread
6. Minute of day
7. ClientCode

The dataset was sorted chronologically and split by date using a 60/20/20 train-validation-test methodology to prevent look-ahead bias.

XGBoost was selected because it handles nonlinear relationships, heterogeneous features, and feature interactions effectively while requiring limited preprocessing. It is also robust to noisy financial datasets and provides strong baseline performance in classification tasks.

The model achieved the following average performance across all horizons:

Training Accuracy: 57.89%

Validation Accuracy: 55.51%

Test Accuracy: 55.63%

The relatively small gap between validation and test performance indicates that the model generalizes reasonably well and is not severely overfitting. Financial market prediction problems are inherently noisy, and predicting short-horizon adverse selection is particularly challenging. Therefore, even modest predictive performance above random chance can have meaningful economic value when incorporated into trading decisions.

The resulting adversity probabilities are subsequently used in Task 4 for externalization decisions and in Task 5 as an input to the dynamic quoting strategy.

Task 4 – Optimal Externalization Threshold

The objective of Task 4 is to determine the optimal probability threshold θ used for externalization. Trades with predicted adversity probability greater than θ are externalized, while the remaining trades are internalized.

The validation-set PnL was evaluated over a range of threshold values between 0 and 1. The resulting PnL-versus-threshold relationship demonstrates the trade-off between over-externalization and under-externalization.

When θ is too low, the strategy externalizes a large fraction of trades. While this reduces exposure to adverse flow, it also removes many profitable trades. Consequently, potential spread capture and trading profits are sacrificed.

When θ is too high, very few trades are externalized. The LP retains profitable flow but also absorbs a significant amount of toxic flow, resulting in losses from adverse selection.

The optimal threshold balances these competing effects by retaining profitable trades while removing sufficiently toxic trades. The results suggest that optimal thresholds differ across clients and horizons. More toxic clients generally benefit from lower thresholds because their trades are more likely to be adverse.

This finding supports the use of client-specific externalization policies rather than a single universal threshold. Client-specific thresholds allow the LP to respond differently to clients with distinct toxicity characteristics.

The final test-set PnL values demonstrate that selective externalization can significantly improve profitability by filtering adverse flow while preserving beneficial trading opportunities.

Task 5 – Dynamic Quoting Under Inventory Pressure

A dynamic quoting framework was developed to adjust bid and ask spreads based on inventory, volatility, predicted adversity, and time remaining in the trading session.

The quoting function takes the form:

$$Q(I, \sigma, \alpha, \eta) \rightarrow (\delta b, \delta a)$$

where I is inventory, σ is realized volatility, α is the adversity probability predicted by the Task 3 model, and η is the elapsed fraction of the trading day.

The spread consists of four components:

1. Volatility adjustment
2. Inventory adjustment
3. Adversity adjustment
4. Time adjustment

Higher volatility leads to wider spreads to compensate for increased price uncertainty. Higher adversity probabilities also widen spreads because informed flow creates adverse-selection risk. Inventory is incorporated through a bounded hyperbolic tangent function, which encourages inventory reduction without generating extreme quote skew. Finally, spreads widen gradually toward the end of the trading day to reduce the probability of carrying inventory overnight.

Model parameters were calibrated on the validation dataset using grid search. The best-performing parameter set was:

$$k_1 = 1.0$$

$$k_2 = 0.002$$

$$k_3 = 0.002$$

$$k_4 = 0.001$$

The calibrated strategy achieved a total test-set PnL of 4,450.12 and a Sharpe-like score of 15.82.

Overall, the results demonstrate that combining inventory control, volatility awareness, and toxicity estimation leads to a more robust market-making strategy capable of balancing profitability and risk management.